The Simpler Theory of Everything Is Exponentially More Probable

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Abstract

The Simplicity Dominance Principle (SDP) states that among all consistent mathematical structures, those with lower Kolmogorov complexity have exponentially higher measure: P(S) 2^(−K(S)). Each additional bit of specification halves a structure's probability. This has a direct consequence for the search for a Theory of Everything: the TOE must reduce total specification complexity relative to current physics, not increase it. The Standard Model plus General Relativity specifies our physics with approximately 190 bits of framework choices plus 20 to 27 free parameter values. Any candidate TOE that requires more total specification than this is exponentially disfavored. String theory, requiring 155 bits of framework plus 181 compactification parameters, increases specification complexity (Bernstein, 2026g). The TOE should be a short generating rule from which the Standard Model's parameters, gauge structure, and gravitational dynamics all compute. SDP does not predict which rule. It predicts its K is low, and that candidates with lower K are exponentially more probable than candidates with higher K.

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1. The Search Direction

Physics has two successful frameworks: the Standard Model of particle physics and General Relativity. They are mutually incompatible at extreme scales. The search for a Theory of Everything seeks to unify or replace them.

SDP constrains this search. Among all consistent mathematical structures capable of supporting observers, those with lower Kolmogorov complexity have exponentially higher measure. Each bit of specification halves the probability. The TOE should therefore have the lowest total K compatible with reproducing observed physics and supporting observers.

This is not a preference. It is a measure-theoretic prediction. A candidate TOE with K = 100 bits is 2^100 times more probable than one with K = 200 bits. The exponential is unforgiving.

2. What Current Physics Costs

The Standard Model specifies particle physics through a gauge group (SU(3)×SU(2)×U(1)), a Higgs mechanism, three fermion generations, and 19 to 26 free parameters whose values are determined by experiment. General Relativity specifies gravity through the Einstein-Hilbert action and a manifold structure. Together, the framework choices require roughly 190 bits. The 20 to 27 free parameter values (including the cosmological constant) have unknown K: their true specification cost depends on whether a generating rule exists.

This is cheap. Two frameworks that describe everything from quarks to black holes, specified in fewer bits than a short text message. SDP explains why: we observe simple physics because simple physics has exponentially higher measure.

3. What the TOE Must Do

The TOE must reproduce the Standard Model and General Relativity (or their successors) from a specification with equal or lower total K. This means one of three things:

Scenario 1: Parameter formula. The 20-27 free parameters turn out to be computable from a short rule. The TOE is the existing frameworks plus this formula. Total K: ~190 bits framework + formula length. If the formula is shorter than the parameter values, K drops. The frameworks stay. Only the unexplained values are explained.

Scenario 2: Framework replacement. A shorter framework generates both the Standard Model and General Relativity, including parameter values, from a single specification. Total K: the length of the replacement rule. If framework and parameters collapse into one object, K can drop far below 190 bits.

Scenario 3: Framework inflation. A larger framework (more dimensions, more symmetry, more structure) derives the Standard Model and General Relativity at the cost of additional specification (compactification, moduli stabilization (fixing extra-dimension sizes), flux assignments (fields threading through them), brane configurations (higher-dimensional membranes)). Total K: higher than current physics. This increases total specification complexity. String theory exemplifies this scenario, requiring 155 bits of framework plus 181 compactification parameters to recover what SM+GR specifies with ~190 bits plus 20-27 values (Bernstein, 2026g).

SDP exponentially favors Scenarios 1 and 2 over Scenario 3. Higher K is exponentially suppressed. The TOE is smaller than what we have, not larger.

4. The Specification Cost of Unification by Inflation

The intuition behind string theory and similar programs is that unification should explain more. String theory does: it derives particle spectra from geometry. But it explains more by specifying more. The compactification geometry required to recover our physics from 10 or 11 dimensions has 181 parameters (Bernstein, 2026g). The "explanation" of 20-27 numbers costs 181 numbers. Under any parsimony principle, this is not explanatory progress.

The error is equating unification with a single framework. True unification under SDP is a single framework with lower total K. If the unified framework requires more specification than the separate frameworks it replaces, the unification costs more than the separation. Unification that increases total specification complexity has not achieved descriptive parsimony. The argument does not depend on SDP specifically: any information-theoretic parsimony principle that penalizes specification complexity yields the same conclusion.

5. The Historical K-Trend

The history of physics provides independent evidence for the prediction that the TOE has low K. Each major unification reduced total specification complexity:

Pre-Newton: Separate laws for terrestrial motion (Galileo) and celestial motion (Kepler). Two frameworks, two sets of parameters. Total K: sum of both.

Newton (1687): One law ($F = Gm\ m\ /r^2$) unified both. Total K dropped: one framework replacing two, one gravitational constant replacing separate terrestrial and celestial parameters.

Maxwell (1865): Four equations unified electricity, magnetism, and optics. Three phenomena collapsed into one framework. Total K dropped again.

Einstein (1915): General Relativity unified gravity with spacetime geometry. The gravitational constant became a feature of spacetime curvature. Framework K dropped further.

Standard Model (1970s): Three forces (electromagnetic, weak, strong) unified under one gauge framework. The number of independent coupling constants dropped from many to three.

At every step, total K decreased. The pattern has held for four centuries without exception. SDP predicts it continues: the TOE has lower K than SM+GR, not higher.

The one apparent exception, quantum mechanics, is instructive. Quantum mechanics introduced new formalism (Hilbert spaces, operators, the Born rule, that probabilities follow the square of the wave amplitude). But it also eliminated the need for separate classical rules governing atoms, spectra, chemistry, and solid-state physics. The total K including all the phenomena it unified dropped, even though the framework itself looked more complex in isolation. Comparing framework complexity without counting what the framework replaces is an accounting error that SDP corrects.

6. Concrete Scenarios

Scenario 1 example: Suppose the 26 parameters of the Standard Model turn out to satisfy a single generating formula computable in 50 bits. The TOE would be SM+GR (~190 bits framework) plus this 50-bit formula, replacing ~150 bits of unexplained parameter values. Net savings: ~100 bits. The TOE is the existing physics plus a short formula. Nothing changes except the parameters are explained.

Scenario 2 example: Suppose a hypergraph rewriting rule of 30 bits generates spacetime, gauge symmetry, three fermion generations, and all 26 parameter values as emergent properties. The TOE would be 30 bits total, replacing ~340 bits (190 framework + 150 parameters). The savings are exponential under SDP: 2^310 times more probable. This is the Wolfram program's aspiration, though no specific rule has been demonstrated.

Scenario 3 example: String theory. Framework requires 155 bits plus 181 bits of compactification specification to recover SM physics. Total K exceeds SM+GR. The "unification" costs more than what it replaces. SDP exponentially disfavors it (Bernstein, 2026g).

7. Predictions

SDP predicts: - The TOE has low K, far below string theory's and at most comparable to SM+GR's ~190 bits - The TOE either reduces the 20-27 free parameters to a formula (Scenario 1) or replaces the entire framework with something shorter (Scenario 2) - If a candidate TOE requires more total specification than SM+GR, it is exponentially disfavored regardless of its other virtues - The search should proceed toward shorter descriptions, not more elaborate ones

These predictions are falsifiable. If the eventual TOE proves to have high K, irreducibly complex with no short generating description, SDP is challenged. Current physics is consistent with the predictions.

8. Objections

6.1 "Simplicity is subjective"

Kolmogorov complexity is objective: the length of the shortest program on a universal Turing machine. Different UTMs yield K values differing by an additive constant (the invariance theorem). For the comparisons that matter here, the differences between candidates span hundreds of bits. No additive constant changes the ordering. The claim is not that we can compute K exactly. It is that the gap between a 50-bit rule and a 300-bit framework is too large for encoding choices to close.

6.2 "The TOE might be irreducibly complex"

Possible. If the final theory of physics requires thousands of bits of irreducible specification, SDP is challenged. But the entire history of physics has moved toward simpler descriptions: Newton unified terrestrial and celestial mechanics, Maxwell unified electricity and magnetism, Einstein unified space and time, the Standard Model unified three forces. Each step reduced total K. A reversal, where the final step increases K, would break a pattern that has held for four centuries. SDP predicts the pattern continues. If it does not, that is evidence against SDP.

9. Conclusion

The Theory of Everything is smaller than what we have, not larger. SDP exponentially favors lower specification complexity. The TOE reduces the Standard Model's unexplained parameters to a formula, or replaces the entire framework with a shorter rule, or both. Higher K is exponentially suppressed. The search direction is toward simplicity.

References

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